

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
FACULTY OF MANAGEMENT STUDIES (FMS)
BACHELOR OF BUSINESS ADMINISTRATION (BBA PROGRAMME)
SEMESTER – IV, UNIVERSITY EXAMINATION, MAY 2019
BM208 OPERATIONS RESEARCH (OLD)
BM208.1 OPERATIONS RESEARCH (NEW)

Date: 02/05/2019, Thursday

Time: 10.00 AM to 10.30 AM

Total Marks: 15

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Important Instructions:

- All questions are compulsory.
- This is a closed book examination.
- There are two sections in the question paper. Section – I contains Q. No.: 1 which is of 15 marks and Section – II contains total five questions wherein Q. No.: 2 to 5 are of 10 marks each and Q. 6 is a Case study of 15 marks.
- Section – I should be answered in the given question paper and Section – II should be answered in a separate answer book.
- Maximum time allotted for Section – I is 30 minutes. Answer Book for Section – II will be given when the Question Paper for Section – I is returned. However, candidates who complete Section – I earlier than 30 minutes, can collect Answer Book for Section – II immediately after handing over Section – I question paper.
- Figures on the right indicate marks for each question.

SECTION – I

Q.1 Objective Type Questions:

A) Match the following: (i – vi)

15
06

Q. No.:		Probable Answer Keys		Write Your Answer Here
i.	Hungarian Method	a)	Simplex Method	i. -
ii.	Exactly Two Decision Variables	b)	Graphical Method	ii. -
iii.	More than Two Decision Variables	c)	Two – Person Zero- Sum Game	iii. -
iv.	Gain of one is loss of other	d)	Assignment Problem	iv. -
v.	Penalty Based Assignment	e)	Critical Activity	v. -
vi.	Total Float = 0	f)	VAM	vi. -
		g)	Free Float = 1	
		h)	Lowest Cost Assignment	
		i)	Laplace Theorem	
		j)	Operational Models	

B) Multiple-Choice Questions (vii – ix) (Note: Encircle the correct answer)**03**

- vii.** Final solution of transportation problem can be identified using
- a) NWCR
 - b) MODI
 - c) Hungarian
 - d) None of these
- viii.** CPM starts with
- a) Forward Pass
 - b) Backward Pass
 - c) Floating
 - d) None of these
- ix.** Prime objective of assignment problem is to
- a) Minimize the Cost
 - b) Maximize Profit
 - c) Balance Profit and Loss
 - d) None of these

C) Fill-in the Blanks (x – xii) You may choose from the following answers:**03**

Added	Critical Activities	PERT / CPM	Subtracted	Total Float	Free Float
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- x.** Slack Variables are to be _____ when the constraint is of less or equal to type.
- xi.** Activities must not have any delay beyond _____ for project to be completed on time.
- xii.** Combination of all _____ will formulate critical path.

D) True or False (xiii – xv) (Note: Encircle the correct answer)**03**

- xiii.** NWCR is always better than VAM and MMM.
(True) or (False)
- xiv.** Graphical method is only for the minimization problem.
(True) or (False)
- xv.** There is always more than one assignment in each row and column.
(True) or (False)

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Date: 02/05/2019, Thursday

Time: 10.30 AM to 01:00 PM

Total Marks: 55

- Note:** - (i) All questions are compulsory.
(ii) This is a closed book examination.
(iii) There are two sections in the question paper. **Section – I** contains Q. No.: 1 of 15 marks and **Section – II** contains total five questions where in Q. No.: 2 to 5 are of 10 marks each and Q. 6 is a Case study of 15 marks.
(iv) **Section – I** should be answered in the given question paper and **Section – II** should be answered in a separate answer book.
(v) Maximum time allotted for **Section – II** is 150 minutes.
(vi) Figures on the right indicate marks for each question.

SECTION – II

Q.2 Solve following LPP using Graphical Method: **10**

$$\text{Max } Z = 5X_1 + 4X_2$$

Subject to

$$6X_1 + 4X_2 \leq 24$$

$$X_1 + 2X_2 \leq 6$$

$$-X_1 + X_2 \leq 1$$

$$X_2 \leq 2$$

$$X_1, X_2 \geq 0$$

OR

Q.2 Solve following LPP using Graphical Method: **10**

$$\text{Max } Z = 15X_1 + 10X_2$$

Subject to

$$4X_1 + 6X_2 \leq 360$$

$$3X_1 \leq 180$$

$$5X_2 \leq 200$$

$$X_1, X_2 \geq 0$$

Q.3 Solve following Transportation Problem to find initial feasible solution using NWCR, LCM and VAM. **10**

		Warehouse				
		A	B	C	D	Supply
Factory	1	19	30	50	10	7
	2	70	30	40	60	9
	3	40	8	70	20	18
	Demand	5	8	7	14	34

Q.4 [A] Identify value of game / saddle point and optimal strategy for following pay-off matrix: [5] 10

Union	Company			
	C1	C2	C3	C4
U1	20	15	12	35
U2	25	14	8	10
U3	40	2	10	5
U4	-5	4	11	0

[B] Identify value of game / saddle point and optimal strategy for following pay-off matrix: [5]

Player A	Player B				
	B1	B2	B3	B4	B5
A1	8	10	-3	-8	-12
A2	3	6	0	6	12
A3	7	5	-2	-8	17
A4	-11	12	-10	10	20
A5	-7	0	0	6	2

OR

Q.4 A small maintenance project consists of following jobs, whose precedence relationships are given below. 10

Activity	A	B	C	D	E	F
Job	1-2	1-3	3-4	2-4	3-5	4-5
Duration (Weeks)	5	12	5	14	15	4

Draw Network Diagram, Calculate Total Float for each activity and identify critical path with total project duration.

Q.5 [A] Solve following Assignment Problem: [5] 10

	A	B	C	D
I	8	10	17	9
II	3	8	5	6
III	10	12	11	9
IV	6	13	9	7

[B] Solve following Assignment Problem: [5]

	M1	M2	M3	M4	M5
J1	6	12	3	11	15
J2	4	2	7	1	10
J3	8	11	10	7	11
J4	16	19	12	23	21
J5	9	5	7	6	10

Q.6 **Case Study:** 15

Solve following LPP using Simplex Method:

$$\text{Max } Z = 11X_1 + 9X_2$$

Subject to

$$3X_1 + 2X_2 \leq 8$$

$$2X_1 + 3X_2 \leq 7$$

$$X_1, X_2 \geq 0$$